Samit Mohan

AI/ML Engineer
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Experience

Beltech AI | *PyTorch, YOLO, Triton, TensorRT, Docker* June 2025 – Present
AI Scientist Bangalore, Karnataka

- B2G security, surveillance, traffic violation and detection organisation working with over **60k** cameras, **27+** cities.
- Designed and deployed an end-to-end vision system over live RTSP streams across **10+ detection modules** (LPR, phone, helmet, side-view-mirror, seatbelt, vehicle make-model-color, no-parking, wrong-way) at **0.9+ F1** for Indian CCTV infrastructure; managed GPU per-card placement and pod autoscaling to prevent OOM under load
- Led the Bangalore on-ground demo end-to-end with a 3-person team: picked junctions, set up and routed camera streams, trained models, built the pipeline, and managed annotators reviewing challans; captured **20k+ violations/day across 6 junctions**
- Developed custom NVIDIA DeepStream pipeline in C keeping decode → detect → track fully on-GPU (no CPU↔GPU copies), scaling per-server capacity to **100+ cameras** on single GPU
- Cut detection latency by **54%** to ≤10 ms on edge (NMS-free YOLOv10 on Triton/TensorRT, 0.90+ mAP across 8 classes); ran an agent-driven input-size sweep cutting weights to 1.2 MB and inference to **30 ms/frame** across all models
- Rebuilt LPR end-to-end (YOLOv11s text detection + PARSeq ViT recognition, TensorRT FP16 on edge), cutting latency **150 ms → 40 ms** with higher read accuracy
- Developed a continuous learning pipeline that ingests misclassifications from live traffic on a 2-week cadence, retrains, validates via mAP50 quality gate, and hot-swaps models on Triton with **zero downtime**
- Designed and deployed a Triton model to manage 7000 fps with PaddleOCR with throughput of 20000 rps along with PUCC FastAPI service (YOLO probe-insertion check + LPR + testing-center OCR vs government registry)

MakeMyTrip | *Scala, Redis, MySQL, Kafka, AWS* August 2022 – October 2022
Machine Learning Intern Gurgaon, Haryana

- Researched on **multi-armed bandit algorithms** (epsilon-greedy, Thompson Sampling, UCB) to optimize homepage engagement; benchmarked strategies on RLHF reward models to rank itinerary recommendations
- Developed an NLP-based resume screening system using text similarity and keyword extraction for Data team hiring

Technical Skills

ML/DL: NumPy, ONNX, PyTorch, YOLO, DeepSORT, ViT, CNNs, Transformers, OpenCV, Fine-tuning, Scikit-learn
LLM / GenAI: RAG, vLLM, embeddings, agents, KV cache, Quantization
Infrastructure: Triton Inference Server, TensorRT, MetaFlow, MLflow, Docker, Kubernetes, Grafana, CI/CD, FFmpeg
Languages & Tools: Python, C/C++, SQL, FastAPI, gRPC, Redis, FAISS, Linux, Git

Projects

MiniTorch | *Python, NumPy, CuPy*

- Built a **PyTorch-like deep learning framework from scratch** with reverse-mode autodiff, topological-sort computation graphs, and optional GPU acceleration via CuPy
- Implemented full Module system (Linear, Conv2d, BatchNorm, Dropout, MaxPool2d, Adam/SGD, LR schedulers) in ∼1000 lines; reached **96%+ on MNIST**, validated via numerical gradient checks

MiniCode | *Python*

- Built a minimal **coding agent loop in ∼250 lines** – tool calls, shell execution, and streaming – to understand what is inside an agentic harness
- Implemented the full tool layer (read, write, edit, glob, grep, shell) without any library import along with an agentic loop that re-invokes the model until tool calls drain, against an **Anthropic-compatible messages API** (Anthropic or OpenRouter)

Teaching & Writing

aiengg.dev | *AI Cohort TA* Dec 2025 – May 2026

- TA'd a 16-week live AI engineering cohort: ran weekly classes on RAG, agents, fine-tuning, and inference for **200+ engineers**; designed assignments, ran doubt sessions, and built the course and syllabus
- InterviewReady: restructured content for **10k+ subscribers**; technical posts reaching **20,000+ readers**

TensorTonic | *ML Practice Platform* Ongoing

- Contributed **300+ ML implementation problems**, system-design problems, interactive quizzes, paper-implementation questions, and the pro study plan for a platform serving **40k+** users

Education

VTU, NHCE 2020 – 2024
B.Tech in Information Technology **(CGPA: 9.4/10.0)** Bangalore, Karnataka

- Founder: Coding Club NHCE